

YI SHI

PhD student

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🔗 <https://yi-shi94.github.io/>

EDUCATION

Doctor of Philosophy (Ph.D.), Computer Science

Simon Fraser University

📅 09/2023 - 06/2027

Master of Science (M.Sc.), Computer Science

New York University

📅 09/2018 - 12/2020

Bachelor of Science (B.Sc.), Computer engineering (Dual Degree Program)

Politecnico di Torino

📅 09/2016 - 12/2017

Bachelor of Engineering (B.Eng.), Electronic and Information Engineering

Tongji University

📅 09/2013 - 12/2017

INDUSTRY EXPERIENCE

Research Intern

Shanghai AI Laboratory

📅 10/2022 - 09/2023 📍 Shanghai, China

Worked on leveraging generative models to enhance 3D character animation generation in both video games and movies.

Supervisor: Jingbo Wang, Bo Dai

- Implemented a framework to evaluate different generative models (DDPM, VAE) to model character motions.
- Researched and implemented diffusion based auto-regressive motion model for enhancing animation realism and achieving real-time interactive control.
- Collaborated with animators and developers to integrate generated character animations into existing workflows of a mobile game.

Machine Learning Engineer

XMOV Inc

📅 09/2020 - 10/2022 📍 Shanghai, China

Created machine learning-driven systems to improve the performance of autonomous 3D avatars capable of supporting natural, real-time online conversations.

Supervisor: Congyi Wang, Jinxiang Chai

- Developed a marker-based motion capture solver for cleaning raw MoCap data. The system is capable of segmenting markers, and predict the position of missing markers. The approach significantly alleviates the workload of data annotators on fixing noisy marker data.
- Implemented a retrieval based chatbot by using fine-tuned a lightweight SentenceBERT for enhanced semantic matching and a fast question pipeline.
- Created a Text-to-Gesture system for generating motions for avatars. Developed a phrase-level language embedding matching system for matching keywords from user speech, and built a system helping users to define custom keypose-to-phrase dictionary.
- Participated in the development of a multilingual Text-to-Speech system. In charge of training a Chinese polyphone (pronunciation of characters) disambiguation model by fine-tuning an Electra Language Model.

Embedded Systems Engineer

STMicroelectronics

📅 08/2017 - 05/2018 📍 Agrate Brianza, Italy

Developed middlewares for a microphone array

- Implemented and optimized real-time acoustic source localization algorithms.

RESEARCH EXPERIENCE

Research Assistant

Simon Fraser University

09/2023 - 06/2027 Burnaby, Canada

Mellow Group, GruVi Lab

Advisor: Xue Bin (Jason) Peng

Character Animation, Generative Model, Reinforcement Learning

- Exploring techniques to train diffusion models with Reinforcement Learning in an end-to-end manner for motor control.
- Developing methods for repurposing pre-trained controllers for downstream tasks.
- Developed methods for training a real-time auto-regressive kinematics-based motion model with DDPM, and steering the denoising process of DDPM with Reinforcement Learning to achieve interactive character control.

Research Assistant

New York University

08/2019 - 05/2020 New York, United States

Multimedia and Visual Computing Lab

Advisor: Yi Fang

Point Cloud Processing, Self-Supervised Learning

- Explored methods for learning expressive embeddings of geometries in a self-supervised manner. Trained a neural network tasked with removing randomly added Gaussian noise on unannotated point cloud data.
- Explored methods for point cloud registration using unsupervised learning and optimization with neural networks.

PUBLICATIONS

Interactive Character Control with Auto-Regressive Motion Diffusion Models

ACM Transactions on Graphics (TOG) / SIGGRAPH 2024

Yi Shi, Jingbo Wang, Xuekun Jiang, Bingkun Lin, Bo Dai, Xue Bin Peng

08/2024 https://yi-shi94.github.io/amdm_page

Polyphone Disambiguation in Mandarin Chinese with Semi-Supervised Learning

Interspeech 2021

Yi Shi, Congyi Wang, Yu Chen, Bin Wang

01/2021 <https://github.com/PaperMechanica/SemiPPL>

Unsupervised Deep Shape Descriptor with Point Distribution Learning

IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR) 2020

Yi Shi, Mengchen Xu, Shuaihang Yuan, Yi Fang

08/2020 <https://github.com/Yi-Shi94/Unsupervised-Deep-Shape-Descriptor-with-Point-Distribution-Learning>

ACADEMIC SERVICES - REVIEWER

CVPR 2023-2024

ICRA 2022-2024

AAAI 2024

ICPR 2024

TOG